Prepare and process data with Azure Databricks

4400 XP

4 hr 1 min remaining • Learning Path0 of 4 modules completed

Level
Intermediate

Product
Azure Databricks

Role
Data Engineer

Subject
Data engineering

Master the essential skills to build robust, scalable data engineering solutions with Azure Databricks and Unity Catalog. Learn to design effective data models, ingest data from diverse sources, transform raw data into analytics-ready formats, and ensure data quality across your lakehouse architecture.

In this learning path, you'll learn how to build a data engineering workflow using Azure Databricks and Unity Catalog. Starting with foundational data modeling concepts, you'll design schemas and partitioning strategies optimized for analytical workloads. You'll then explore multiple ingestion patterns—from managed connectors to streaming pipelines—to bring data into your lakehouse. Next, you'll apply transformation techniques to cleanse and reshape data for business use. Finally, you'll implement quality controls to maintain data integrity throughout your pipelines. By the end, you'll have the practical skills to architect and build production-ready data solutions in Unity Catalog.

Prerequisites

- Good understanding of Azure Databricks workspaces and Unity Catalog concepts
- Familiarity with SQL and Python programming
- Knowledge of fundamental data engineering and data warehouse concepts

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Redeem your code

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Modules in this learning path

Design and implement data modeling with Azure Databricks

1400 XP

1 hr 32 min remaining • Module0 of 13 units completed

Effective data modeling forms the foundation of a performant and maintainable data platform. This module explores how to design ingestion logic, select appropriate tools and table formats, implement partitioning schemes, manage slowly changing dimensions, choose appropriate data granularity, and optimize table performance through clustering strategies in Azure Databricks with Unity Catalog.

Start >

Overview

Introduction
3 min

Design ingestion logic and data source configuration
10 min

Choose a data ingestion tool
11 min

Choose a data table format
8 min

Design and implement a data partitioning scheme
10 min

Choose a slowly changing dimension (SCD) type
8 min

Implement a slowly changing dimension (SCD) type 2
4 min

Design and implement a temporal (history) table to record changes over time
6 min

Choose granularity on a column or table based on requirements
9 min

Choose managed vs unmanaged tables
7 min

Design and implement a clustering strategy
8 min

Knowledge check
5 min

Summary
3 min

Ingest data into Unity Catalog

1100 XP

56 min remaining • Module0 of 10 units completed

Data ingestion is a fundamental capability for any data platform. This module explores the comprehensive set of techniques available in Azure Databricks for loading data into Unity Catalog tables. You'll learn how to use managed connectors with Lakeflow Connect, write custom ingestion code in notebooks, apply SQL commands for batch file loading, process change data capture feeds, configure streaming ingestion from message buses, set up Auto Loader for automatic file detection, and orchestrate ingestion workflows with Lakeflow Spark Declarative Pipelines.

Overview

Introduction
2 min

Ingest data with Lakeflow Connect
6 min

Ingest data with notebooks
7 min

Ingest data with SQL methods
7 min

Ingest data with CDC feed
5 min

Ingest data with Spark Structured Streaming
8 min

Ingest data with Auto Loader
6 min

Ingest data with Lakeflow Spark Declarative Pipelines
8 min

Module assessment
5 min

Summary
2 min

Cleanse, transform, and load data into Unity Catalog

1100 XP

55 min remaining • Module1 of 10 units completed

Data engineering requires transforming raw data into clean, well-structured formats ready for analysis. This module explores techniques for profiling data quality, selecting appropriate column types, resolving duplicates and null values, applying filtering and aggregation transformations, combining datasets with joins and set operators, reshaping data through pivoting and denormalization, and loading transformed data using append, overwrite, and merge strategies.

Continue

Overview

Introduction
2 min

Profile data
6 min

Choose column data types
8 min

Resolve duplicates and nulls
7 min

Transform data with filters and aggregations
7 min

Transform data with joins and set operators
7 min

Transform data with denormalization and pivots
7 min

Load data with merge, insert, and append
6 min

Module assessment
5 min

Summary
2 min

Implement and manage data quality constraints with Azure Databricks

800 XP

38 min remaining • Module0 of 7 units completed

This module explores strategies for maintaining high data quality in Azure Databricks. You will learn how to implement validation checks, enforce schemas, manage schema drift, and use pipeline expectations to ensure data integrity throughout your data pipelines.

Overview

Introduction
2 min

Implement validation checks
8 min

Implement data type checks
7 min

Detect and manage schema drift
8 min

Manage data quality with pipeline expectations
6 min

Module assessment
5 min

Summary
2 min